Sensor applications

Characterizing the Dynamics of Surface Electromyography Signals in Muscle Fatigue Through Visibility Motif Networks

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Abstract—Monitoring of the physiological function during exercise can provide insights on the quality of the training and prevent injury. Specifically, the signals from the muscle sensors (surface electromyography) are difficult to interpret and limited attempts have been made to develop effective algorithms for the real-time monitoring of muscle fatigue. In this work, the applicability of visibility graph motif features for the real-time monitoring of muscle fatigue is explored. Experimental investigations have been conducted on 58 healthy adult volunteers. Results indicate that the network entropy features are able to characterize the changes in signal dynamics in nonfatigue and fatigue conditions. These metrics have the potential to be used as a marker to predict functional capabilities of humans in real-world scenarios.

Index Terms—Sensor applications, muscle fatigue, surface electromyography, transition network, visibility graph motif.

I. INTRODUCTION

Surface electromyography (sEMG) is a technique used to sense the electrical activity of the muscles. This sensing scheme utilizes metal electrodes to convert ionic potentials generated during contraction to electric voltage. These electric potentials allow us to characterize the underlying physiological status of the muscle [1]. One key application of this method is to detect exercise induced muscle fatigue [2].

The sEMG signals are difficult to interpret using conventional signal processing methods due to the nonlinear signal variations. Some of the previous attempts include the use of features such as root mean square value, median frequency, bispectral index, and time-frequency index [3]. These techniques allow for offline assessment of signal [4].

Recently, we introduced techniques such as multiscale entropy [4], visibility graph motifs [5], and symbolic representation methods [6] to quantify muscle fatigue. In particular, the transition network feature obtained from the symbolic representation of the signal resulted in good classification performance without the need of complex machine learning algorithms such as SVM and neural network [7].

In this study, a new approach to quantify the dynamics of the signal has been proposed by combining visibility graph motif with transition networks.

II. METHODS

A. Signal Acquisition

Experimental dynamic muscle fatigue investigations have been conducted on 58 healthy adult volunteers under controlled condition.

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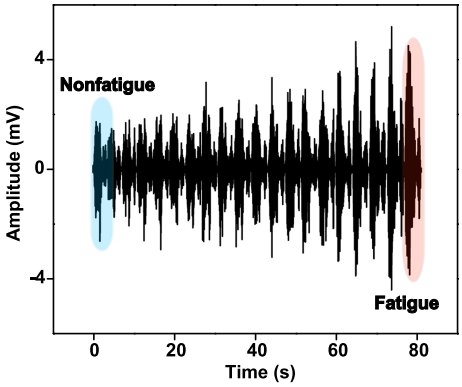

Fig. 1. Representative sEMG signal during dynamic contraction exercise.

The experimental protocol has been approved by the IIT Madras Institutional Review Board (IEC/2017/04/SR-2/02). sEMG signals are recorded from these participants at a sampling rate of 10 000 Hz using bipolar surface electrodes placed on the biceps brachii muscle. More information on this is found in Makaram et al. [7].

B. Preprocessing

The acquired signals are downsampled to 1000 Hz, filtered using a 10–400 Hz bandpass filter and a 50-Hz notch filter. The dynamic curls are segmented based on the algorithm given in Makaram et al. [7]. The first and last curl of the signal is considered as nonfatigue and fatigue condition, respectively.

C. Signal Analysis

Visibility graph motifs are constructed from the nonfatigue and fatigue segments based on the work of Navaneethakrishna and Ramakrishnan [5]. A 1-D Markov Model is constructed to analyze the dynamics of the visibility graph motifs over time. Transition network

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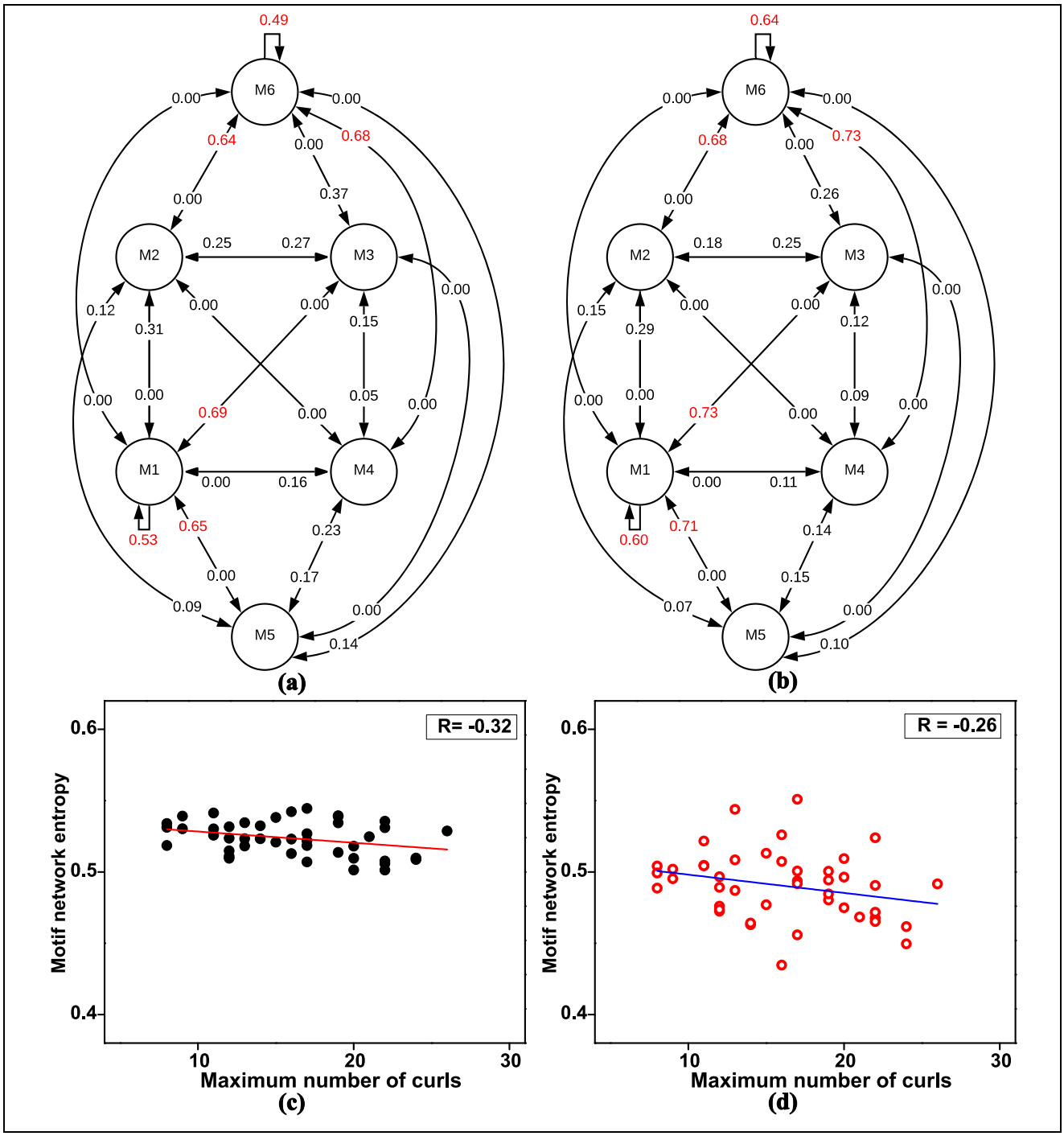

Fig. 2. Visibility graph motif transition network in (a) nonfatigue and (b) fatigue condition for a representative subject and the variation of the network entropy features for subjects with different maximum number of curls in (c) nonfatigue and (d) fatigue condition.

entropy feature is extracted from the constructed network. Details on the transition network entropy are given in Makaram et al. [7]

$$\text{Network Entropy} = \frac{1}{N_s} \sum_i \frac{\sum_j p_{ij} \times \log_2 p_{ij}}{\log_2(N_s)} \tag{1}$$

where N_s represents the number of states ($N_s = 6$), and p_{ij} is the transition probability for going from state i to state j.

III. RESULTS AND DISCUSSION

Fig. 1 shows a representative sEMG signals recorded from biceps brachii during the dynamic contraction exercise. The signals from all the participants are found to be complex and random. The repetitive flexion and extension action of the elbow joints during the biceps curl exercise is the reason for the periodic burst like activity of the signals. In flexion movements, the biceps act as the agonist muscle and triceps act as the antagonist muscle. On the other hand, during extension, the triceps act as the agonist muscle and biceps act as the antagonist muscle.

In Fig. 2(a), the motif network obtained during dynamic contraction in nonfatigue condition is shown. M1 to M6 corresponds to different visibility graph motifs configurations with 4 nodes. M1 and M6 have higher self-transition probabilities with values 0.53 and 0.49, respectively. Additionally, the transition to states M3-M1, M5-M1, M2-M6, and M4-M6 are 0.69, 0.65, 0.64, and 0.68, respectively.

In Fig. 2(b), the motif network obtained during dynamic contraction in fatigue condition is shown. The M1 and M6 have higher self-transition probabilities with values 0.60 and 0.64, respectively.

Additionally, the transition to states M3-M1, M5-M1, M2-M6, and M4-M6 are 0.71, 0.73, 0.68, and 0.73, respectively. In both these cases, the self-transition probabilities and probabilities of transition to motifs M1 and M6 are higher in fatigue condition. This increased probability indicates the presence of self-similar patterns.

The correlation between maximum number of curls performed and the motif network entropy in nonfatigue and fatigue conditions is shown in Fig. 2(c) and (d). It is seen that a negative correlation of $R = -0.32$ is observed in the nonfatigue condition. This correlation is statistically significant with $p < 0.05$. This decreasing trend in the entropy value might be due to the differences in the motor unit recruitment pattern of the muscle. In the fatigue case, a decreasing trend is observed with increased number of curls. However, the R-value of -0.26 observed is not statistically significant.

In comparison, fatigue condition exhibits lower entropy values indicating reduced complexity. This shows that the signal obtained is more regular and exhibiting self-similar patterns in fatigue as seen in other varied biological conditions. The nonfatigue condition exhibits lower intersubject variability. The feature is statistically significant in differentiating nonfatigue and fatigue conditions with $p < 0.005$.

IV. CONCLUSION

Reliable sensing and estimation of muscle fatigue is a complex task due to the nonlinear signal variations in surface electromyography signals. The signals are influenced by subject specific physiological characteristics such as the anthropometry, muscle fiber distribution, and neural control. In this study, experimental investigations have been conducted to characterize muscle fatigue in age matched subject population with an aim to discern the variation in signal dynamics and differentiate fatigue condition. The results indicate that the visibility motif transition networks can differentiate nonfatigue and fatigue conditions, and lower entropy values in fatigue indicate more self-similar patterns. The regression analysis shows the predictive capabilities of the feature to determine physical limits of the subject. It appears that the proposed pipeline would be beneficial in the exercise physiology and sports training fields.

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